Dynamic Predictive Model for Growth of *Salmonella* Enteritidis in Egg Yolk

Gompertz Model with Modified Ratkowsky Secondary Model

Boyuan Zhao, Wenyu Xu, Yizhou Li

College of Biosystems Engineering and Food Science
Zhejiang University

April 21, 2026

Outline

1 Introduction

2 Model

3 Data & Parameters

4 Forward Problem

5 OLS Estimation

6 Residual Analysis

7 Optimal Experimental Design

8 Bootstrap

9 Summary

10 References

Introduction & Background

- *Salmonella* Enteritidis (SE) — leading cause of foodborne illness
- Shell eggs: SE penetrates shell → grows in iron-rich yolk
- USDA-FSIS: eggs stored $\leq 7.2\,^{\circ}\mathrm{C}$ within 12 h of laying
- Temperature fluctuates during cooling, storage, distribution
- **Objective:** Develop a dynamic Gompertz model for SE growth under sinusoidal temperature (3–43 °C)

Ref: Gumudavelli et al. (2007), *J. Food Sci.* 72:M254–62

Salmonella bacteria (colorized SEM).
Source: NIAID/CDC, public domain.

Model Description

Primary Model (Gompertz)

Constant T (analytical):

$$y(t) = A + C \cdot e^{-e^{(\mu e/C)(M-t)+1}}$$

Variable T (ODE):

$$\frac{dy}{dt} = -\mu(T)\, e\, \frac{y-A}{C}\, \ln\left(\frac{y-A}{C}\right)$$

$y = \log_{10} N$; A: initial; C: range; M: inflection time

Secondary Model (Ratkowsky)

$$\mu(T) = a(T - T_{\min})^2 \left(1 - e^{b(T - T_{\max})}\right)$$

a, b: regression coefficients

$T_{\min} = 6\,°\mathrm{C}$, $T_{\max} = 46.3\,°\mathrm{C}$ (fixed)

Dynamic Model

Integrate primary + secondary, solve numerically with `ode45`.

Temperature interpolated via `interp1`.

Data & Parameters

Data Summary

- Growth: 12 time points (0–17 hr), 20 observations
- 0–2 hr: single; 3–17 hr: duplicates
- Temperature: 47 points, sinusoidal ∼7–43 °C
- Interpolated via `interp1`

Param	Description	Guess	Status
A	Init. $\log_{10}$(CFU/mL)	2.60	Est.
C	Growth range	11	Est.
M	Inflection time (hr)	7.5	Est.
a	Ratkowsky coeff	3.38e-4	Est.
b	Ratkowsky coeff	0.275	Est.
$T_{\min}$	Min growth temp	6.0 °C	Fixed
$T_{\max}$	Max growth temp	46.3 °C	Fixed

Table 1. Model parameters and initial guesses.

Forward Problem & Scaled Sensitivity Coefficients

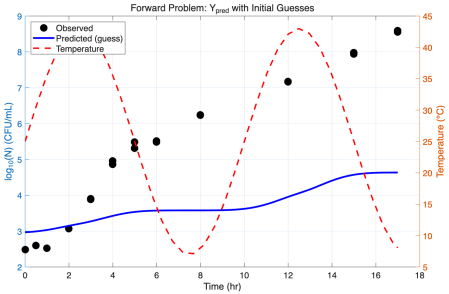

Fig. 1. Forward prediction with initial guesses.

Black: observed; Blue: predicted; Red: temperature.

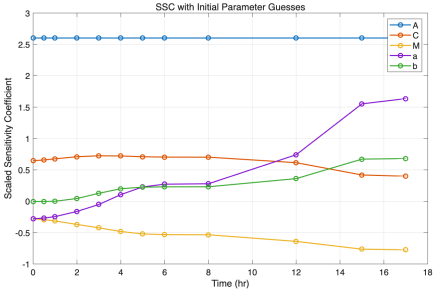

Fig. 2. Scaled sensitivity coefficients (SSC) at initial guesses.

All 5 parameters show distinct patterns.

$T_{\min}$, $T_{\max}$ have negligible SSC $\rightarrow$ fixed at literature values.

OLS Parameter Estimates

Param	Estimate	SE	Rel.Err (%)	$\mathrm{CI}_{\mathrm{lo}}$	$\mathrm{CI}_{\mathrm{hi}}$
A	2.4011	0.2540	10.58	1.8597	2.9425
C	6.3798	0.4565	7.16	5.4067	7.3528
M	2.9213	1.8074	61.87	−0.9310	6.7736
a	0.001135	0.000263	23.15	0.000575	0.001695
b	0.2621	0.2180	83.20	−0.2027	0.7268

Table 2. OLS parameter estimates with standard errors and 95% confidence intervals.

	A	C	M	a	b
A	1.000	−0.834	0.808	−0.281	0.459
C		1.000	−0.711	0.050	−0.312
M			1.000	−0.601	0.826
a				1.000	−0.930
b					1.000

Table 3. Correlation matrix.

Metric	Value
RMSE	0.2524
Pseudo-R^2	0.9874
MSE	0.0637
N	20

Table 4. Goodness of fit.

Fitted Curve with Confidence & Prediction Bands

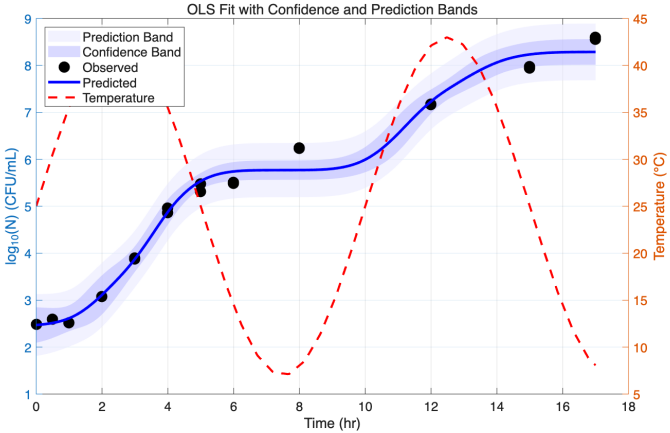

Fig. 3. OLS fitted curve with 95% asymptotic confidence band (dark) and prediction band (light).

Black dots: observed; Blue line: predicted; Red dashed: sinusoidal temperature.

Residual Analysis

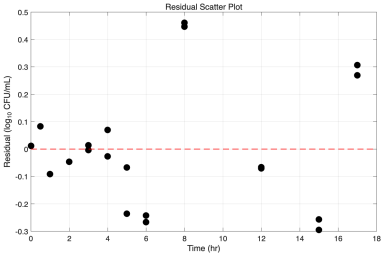

Fig. 4. Residuals vs. time.

Random scatter around zero.

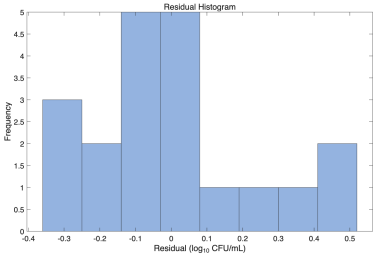

Fig. 5. Residual histogram.

Approximately bell-shaped.

#	Assumption	Result
1	Model is correct	✓ (visual)
2	Errors are random	DW $= 1.3149$
3	Constant variance	✓ (visual)
4	Errors uncorrelated	See DW
5	Normal distribution	SW $p = 1.0000$ (✓)

Table 5. Five standard statistical assumptions.

Final Scaled Sensitivity Coefficients

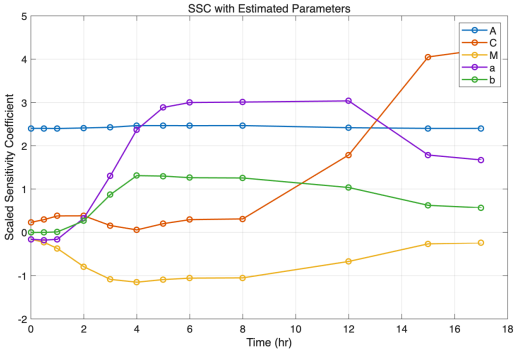

Fig. 6. SSC recomputed with estimated parameters. Patterns consistent with initial SSC.

Param	max|SSC|
A	2.4649
C	4.2172
M	1.1521
a	3.0349
b	1.3092

Table 6. Final SSC magnitudes.

All parameters identifiable at the OLS solution.

Optimal Experimental Design

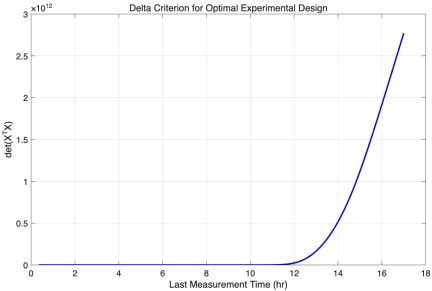

Fig. 7. Delta criterion $\det(X^T X)$ vs. last measurement time. Increasing trend: more data improves estimation.

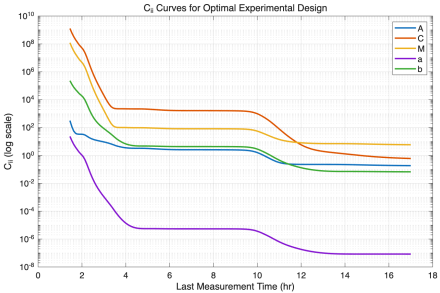

Fig. 8. C_{ii} curves (log scale). Decreasing values: parameters become more precise.

Key measurement window: exponential growth phase ($\sim$2–8 hr).

Bootstrap Analysis ($N = 1000$)

	Bootstrap CI		Asymptotic CI	
Param	Lower	Upper	Lower	Upper
A	1.600	2.741	1.860	2.943
C	5.761	7.819	5.407	7.353
M	-0.624	5.388	-0.931	6.774
a	8.81e-4	2.11e-3	5.75e-4	1.70e-3
b	0.065	—	-0.203	0.727

Table 7. Bootstrap vs. asymptotic 95% CI.

Band	Avg Width
Asymptotic CB	0.5307
Bootstrap CB	0.4058
Asymptotic PB	1.2042
Bootstrap PB	1.0720

Table 8. Band width comparison.

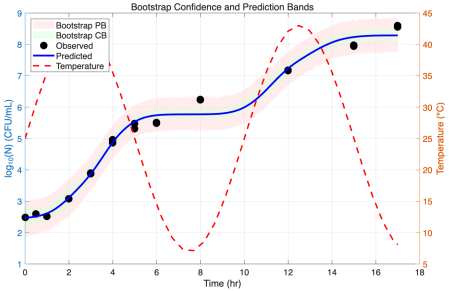

Fig. 9. Bootstrap CB (green) and PB (pink) from 1000 residual-resampling iterations.

Summary & Conclusions

- Dynamic Gompertz model successfully developed for SE growth in egg yolk
- 5 parameters estimated, 2 fixed; RMSE $= 0.2524$ $\log_{10}$ CFU/mL
- Pseudo-$R^2 = 0.9874$; all 5 statistical assumptions evaluated
- Optimal experimental design: key window at 2–8 hr
- Bootstrap CIs generally consistent with asymptotic CIs
- **Practical significance:** Model predicts SE risk during egg cooling, storage, and distribution — supports USDA-FSIS 7.2 °C recommendation

References

Gumudavelli V et al. 2007. Dynamic predictive model for growth of *Salmonella* Enteritidis in egg yolk. *J. Food Sci.* 72:M254–62.

Baranyi J, Roberts TA. 1994. A dynamic approach to predicting bacterial growth in food. *Int J Food Microbiol* 23:277–94.

Zwietering MH et al. 1991. Modeling of bacterial growth as a function of temperature. *Appl Environ Microbiol* 57:1094–101.

Beck JV, Arnold KJ. 1977. *Parameter Estimation in Engineering and Science*. Wiley.

Thank You!

Questions?